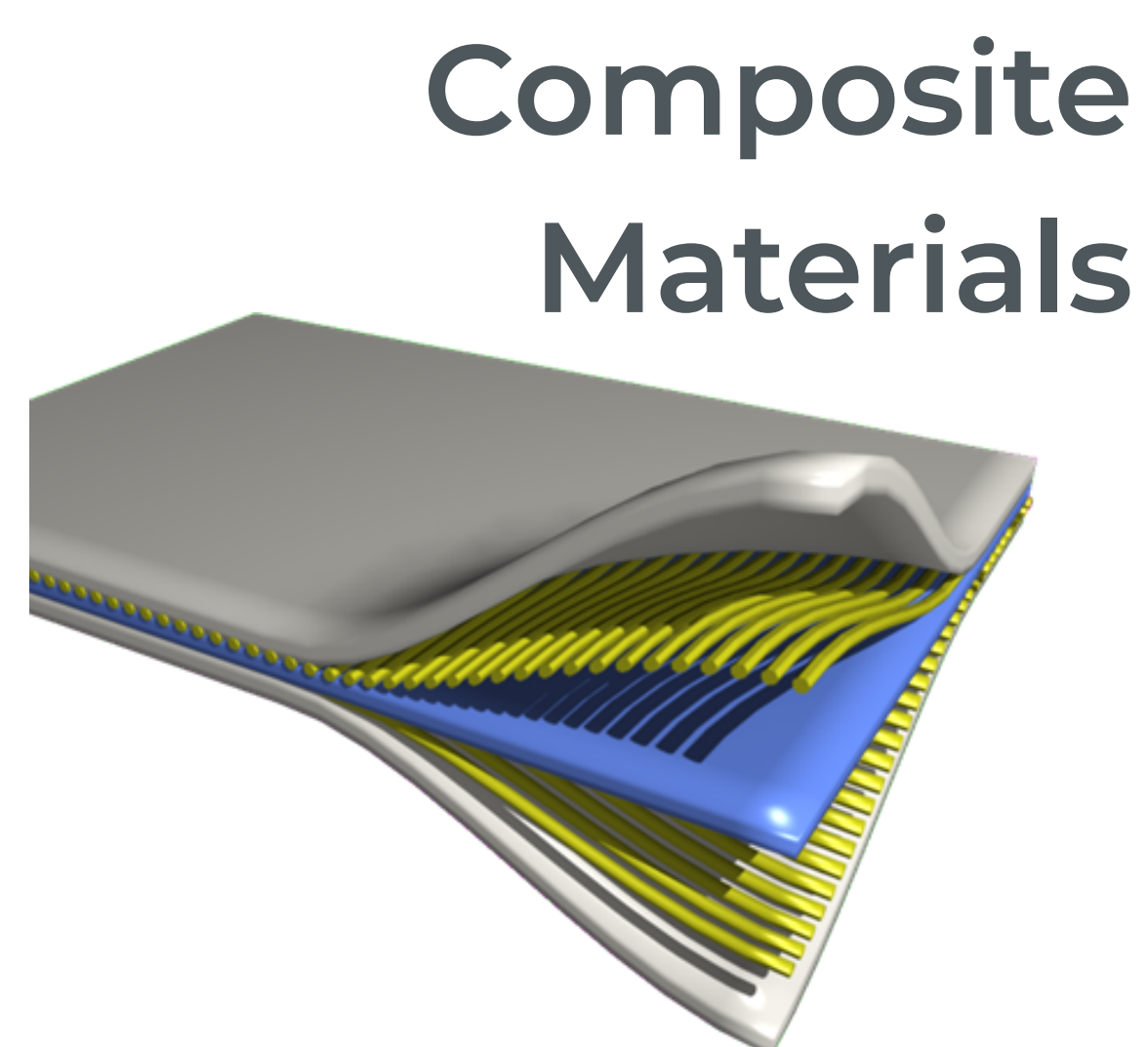


MULTIMODAL OPTIMIZATION OF PIEZOELECTRIC PATCH PLACEMENT AND SHUNT CIRCUIT DESIGN FOR ENHANCED STRUCTURAL VIBRATION CONTROL

Introduction



- ✈️ 80% Structural Volume
50% Structural Weight
- 🚗 87% Battery module mass reduction
- 🤖 28% faster turn rates
50% smaller turn radii
50% faster speed

Problems

Structural Failures
Structural Instabilities
Excessive amplitudes

Solution

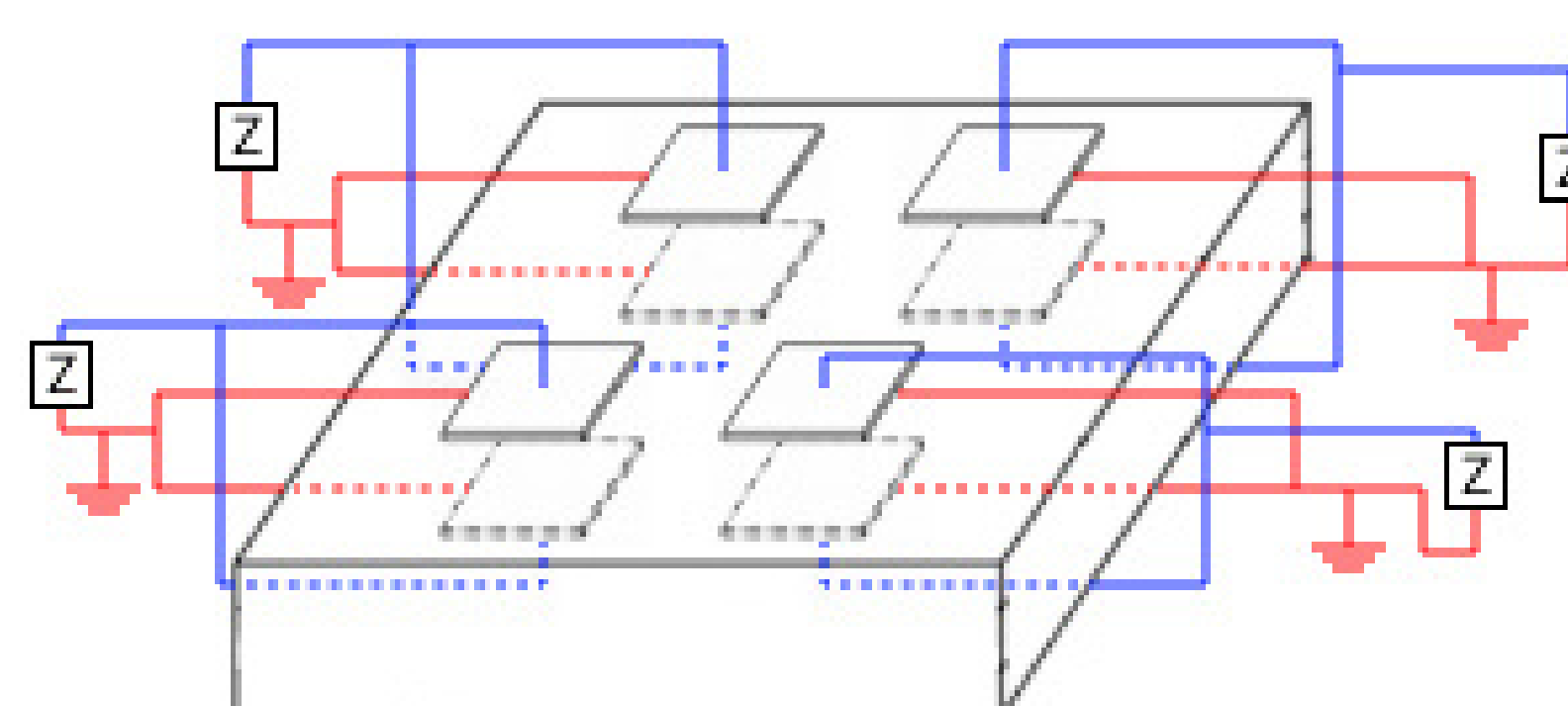
Piezoelectric
Material



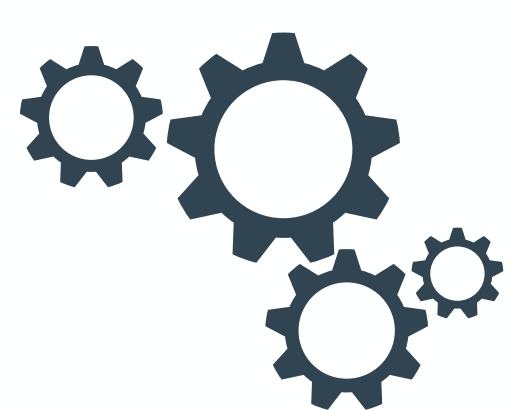
Shunt
Circuit



Smart material with passive shunt control to reduce vibrations by optimizing piezoelectric patches and electrical elements.



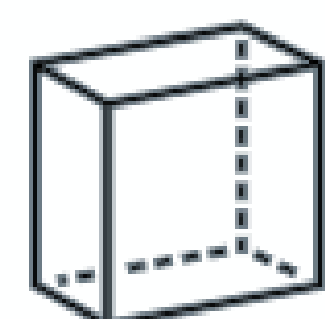
Challenges



Mechanics

Electromechanical coupling coefficient

$$k^2 = \frac{\omega_{OC}^2 - \omega_{SC}^2}{\omega_{SC}^2}$$



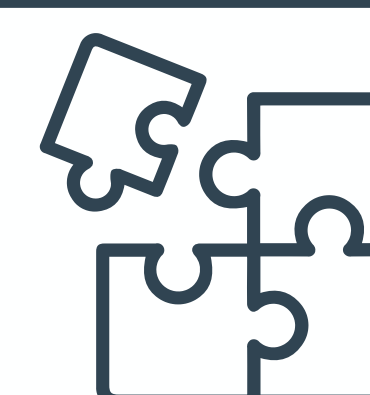
Geometrics

Patch Pairs Positions
Patch Pairs Quantity
Patch Pairs Geometry



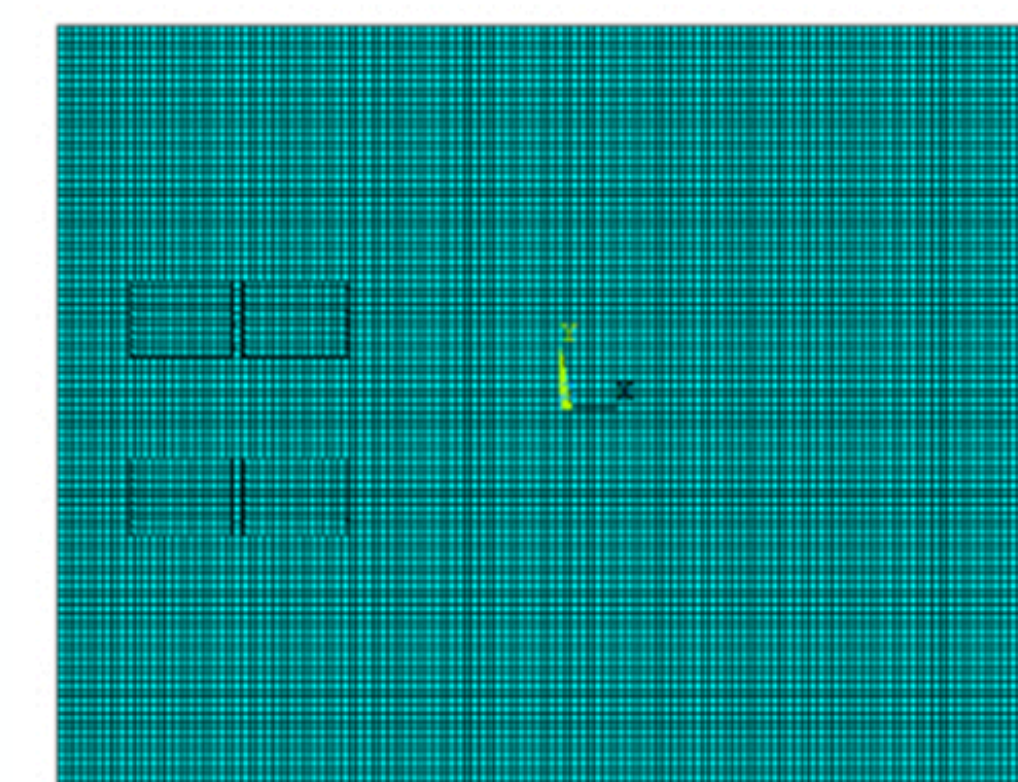
Electrics

Circuit Type
Shunt Type
RL values



Models

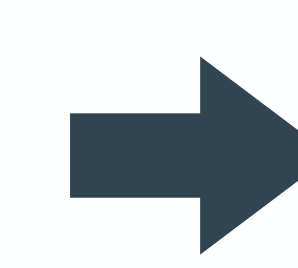
ANSYS



OPTIMIZATION

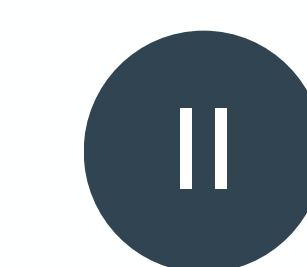


Positions and
Quantities

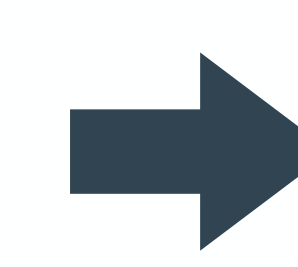


EMCC

GLODS



Resistors and
Inductors



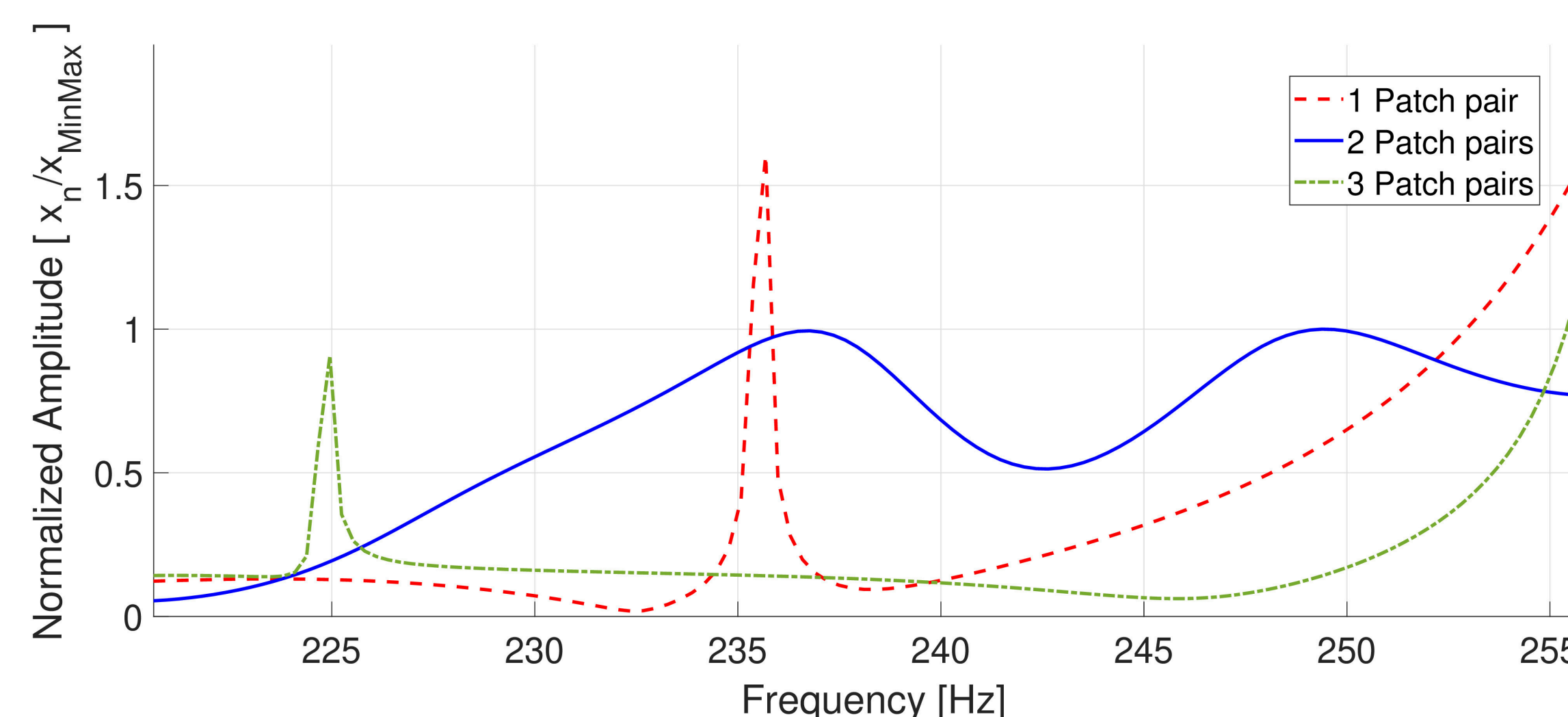
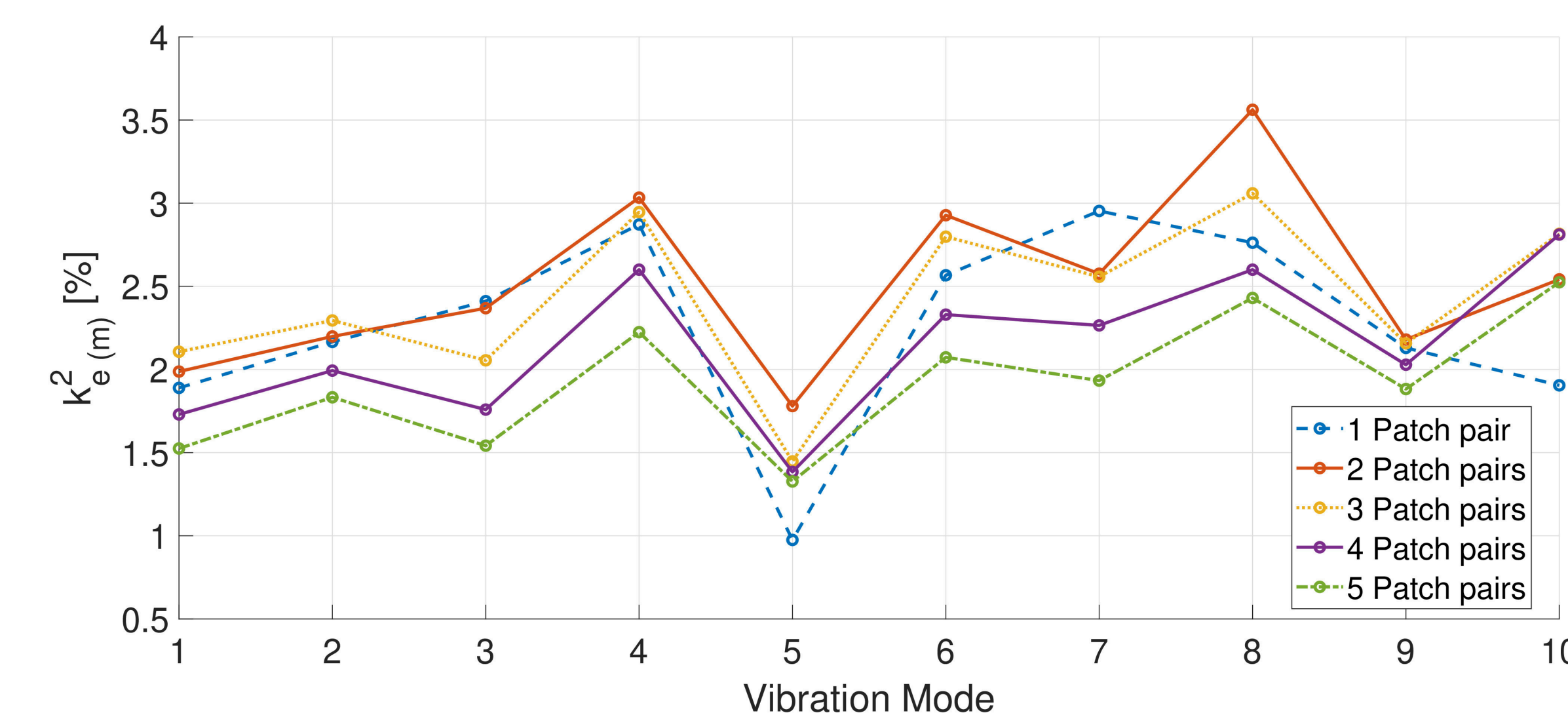
Amplitude

DMS

Results

UNIMODAL APPROACH

Splitting up to 3 patches is efficient
Single shunt isn't good for high frequency modes
1 patch pairs configuration is most efficient for low frequency modes



MULTIMODAL APPROACH

Multishunt circuit is the most efficient for multimodal applications
2 patch pairs was the best result for modes 6, 7 and 8
2 patch pairs was the best result for modes 6, 7 and 8

Conclusion and Next Steps

Low frequency: The best performance occurs with 1 patch pair in unimodal tuning, maximizing EMCC and reducing vibration peaks.

High frequency: Multimodal multi-shunt tuning improves amplitude reduction, but increasing patch division beyond two introduces mass effects that limit performance.

Next steps: Integrate all optimizations to balance patch number, shunt tuning, and mass distribution.